

# Lecture 3

Q: Which properties of quaternion algebras generalize?

Def: A finite dimensional algebra  $A$  over a field  $F$  is called a central simple algebra if the center of  $A$  is exactly  $F$  and  $A$  contains no nontrivial proper two-sided ideals.

There is a beautiful theory of CSAs, but it is beyond the scope of this course. We'll prove one result about quaternion algebras in this context.

Prop: Every quaternion algebra over  $F$  is a central simple algebra.

If you did the exercise from Lecture 1 that showed  $\mathbb{Z}(\mathbb{H}) = \mathbb{R}$ , then some of this will feel familiar!

Proof: Let  $A := (a, b)_F$  be a quaternion algebra w/ standard basis  $\{1, i, j, ij\}$ .

$A$  is central ( $\mathbb{Z}(A) = F$ ): Let  $v = v_0 + v_1 i + v_2 j + v_3 ij \in \mathbb{Z}(A) \subseteq A$ . Then  $v$  commutes with  $j$ , so  $0 = vj - jv = 2(v_1 ij + v_3 bi)$ .

Since  $\text{char } F \neq 2$  and  $i, ij$  are lin. indep. over  $F \Rightarrow v_1 = v_3 = 0$ .

Similarly  $v$  commutes w/  $i$ , so  $0 = vi - iv = 2v_2 ij \Rightarrow v_2 = 0$ .

Thus  $v = v_0 \in F$ . So,  $\mathbb{Z}(A) \subseteq F$  and  $F \subseteq \mathbb{Z}(A) \Rightarrow A$  is central.

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A is simple: We give different proofs depending on whether  $A$  is split.

If  $A$  is a division algebra over  $F$ , then every nonzero elt is invertible. Let  $I \subseteq A$  be a nonzero two-sided ideal and let  $x \in I$  be nonzero. Then  $x^{-1} \in A$  and hence  $1 = x^{-1}x \in I$ . Thus  $I = A$  and  $A$  has no nontrivial proper 2-sided ideals.

If  $A \cong M_2(F)$ , let  $J \subseteq M_2(F)$  be a nonzero 2-sided ideal. The goal is to show  $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in J$ . Take a nonzero  $M = (m_{ij}) \in J$  WLOG we may assume  $m_{11} \neq 0$ .

Since  $J$  is 2-sided and  $E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \in M_2(F)$  we find:

$$E_1 M E_1 = \begin{pmatrix} m_{11} & 0 \\ 0 & 0 \end{pmatrix} \in J.$$

Since  $m_{11} \in F^\times$ , we can scale by  $\begin{pmatrix} m_{11}^{-1} & 0 \\ 0 & m_{11}^{-1} \end{pmatrix}$  to get

$E_1 \in J$ . Similarly, one can show  $E_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \in J$ .

Thus  $J$  contains  $E_1 + E_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow J = M_2(F)$ .

Thus  $A$  is simple.  $\blacksquare$

Now we'll turn our attention to geometry to obtain another condition for detecting splitness.

## The associated conic:

Recall that a quaternion algebra  $A = (a, b)_F$  splits iff there exists  $q \in A$  s.t.  $N(q) = 0$ .

We can make life easier by looking at the 3-dimensional subspace of pure quaternions

$$A^0 := \{xi + yj + zij \in A \mid x, y, z \in F\}$$

If  $q \in A^0$  and  $N(q) = 0$  this says

$$-ax^2 - by^2 + abz^2 = 0$$

$$\Rightarrow ax^2 + by^2 = abz^2$$

By a change of variables and considering solutions w/  $z \neq 0$  we obtain an equation of the form

$$aX^2 + bY^2 = 1. \text{ (See notes for details).}$$

Let  $(a, b)_F$  be a quaternion algebra. The associated conic  $C(a, b)$  is the affine plane curve defined by the equation  $ax^2 + by^2 = 1$ .

We say  $C(a, b)$  has an  $F$ -rational point if there exists a pair  $(x, y) \in F^2$  satisfying this equation.

Let's see how the associated conic gives us a characterization for splitting.

## Theorem (geometric condition for splitness)

For  $a, b \in F^\times$  the following conditions are equivalent:

- 1)  $(a, b)_F \cong M_2(F)$
- 2) The equation  $ax^2 + by^2 = 1$  has a solution  $(x, y) \in F^2$
- 3) The equation  $ax^2 + by^2 = z^2$  has a solution  $(x, y, z) \in F^3$  other than  $(0, 0, 0)$ .

### Proof:

(2)  $\Rightarrow$  (3):  $(x, y)$  a soln to  $ax^2 + by^2 = 1 \Rightarrow (x, y, 1)$  is a soln to  $ax^2 + by^2 = z^2$ .

(3)  $\Rightarrow$  (1): Suppose  $(x, y, z) \neq (0, 0, 0)$  is a soln to  $ax^2 + by^2 = z^2$ .

If  $y = 0$ , then  $x \neq 0$  so  $a = \left(\frac{z}{x}\right)^2$  is a square thus the algebra splits by Thm 2.12 from notes.

If  $y \neq 0$ , rearranging the equation gives

$$b = \left(\frac{z}{y}\right)^2 - a\left(\frac{x}{y}\right)^2 \Rightarrow b \in Na \Rightarrow (a, b)_F$$

splits by Thm 2.12.

(1)  $\Rightarrow$  (2) Suppose  $(a, b)_F \simeq M_2(F)$  so  $(a, b)_F$  is split. By Thm 2.12,  $b \in N_a \Rightarrow b = u^2 - av^2$  for some  $u, v \in F$ . Rearranging gives

$$av^2 + b(1)^2 = u^2.$$

If  $u \neq 0$  then  $a\left(\frac{v}{u}\right)^2 + b\left(\frac{1}{u}\right)^2 = 1$ , i.e.

$\left(\frac{v}{u}, \frac{1}{u}\right) \in F^2$  is a solution to the eqn

$$ax^2 + by^2 = 1.$$

If  $u = 0$ , then  $b = -av^2$  and we leave you to check that  $\left(\frac{1+a}{2a}, \frac{1-a}{2av}\right)$  is a solution.  $\square$

There are many corollaries of this result. The notes contain several. I'll record one here.

Cor: Let  $F = \mathbb{F}_q$  be a finite field. Every quaternion algebra  $(a, b)_F$  over  $F$  is split.